Artificial Intelligence Lab (AI 2002)

Date: March 17th, 2025

Course Instructor(s)
Ms. Ramsha Jat

Lab Mid Exam Paper B

Total Time: 90 minutes

Total Marks: 50

Total Questions: 03

Semester: Spring 2025

Campus: Karachi

Department: AI

Instructions:

- The Exam paper consists of 3 questions on 2 pages.
- Submit ipynb files name under your nu id with question number. E.g: k22-xxxx_Q1.ipynb.
- Submission will be on Software. Submission IP is **172.16.33.88:5000.**
- Enter your NU Id in Username and **Fast1234** in password.
- Read the question carefully before answering.

Student Name	Roll No	Section	Student Signature

LLO # 1: [18 Marks]

Q1. A pirate agent is stranded on a grid-based island and must locate buried treasure. The island's terrain consists of water (W), land (L), rocks (R), and the treasure (T). The pirate can only move on land (L) and must navigate using breadth-First Search (BFS) to find the shortest path to the treasure. The agent's goal is to reach the treasure (T), and it cannot traverse water (W) or rock (R) cells. Identify the type of agent and implement BFS would be implemented to solve this problem. For the example input map below, show the path taken to reach the treasure.

```
L L L W T
L R L W W
L L L L W
R R L R W
L L L L L
```

LLO # 3: [18 Marks]

Q2. Implement an A search algorithm* to find the shortest path in an N x M weighted grid, where each cell represents a terrain cost (a positive integer). Some cells are blocked (#), meaning they

cannot be traversed. The algorithm should determine the least-cost path from the top-left corner (0,0) to the bottom-right corner (N-1, M-1) while considering the terrain costs. The movement is restricted to four directions: up, down, left, and right, and the cost of movement is determined by the value of the destination cell. The heuristic function should use Manhattan distance to estimate the remaining cost to the goal. The implementation should efficiently handle larger grids and return both the optimal path and total travel cost.

1	1	2	#	3
1	#	2	#	#
1	2	1	2	#
#	#	1	#	#
1	1	1	2	1

LLO # 4: [14 Marks]

Q3. Implement a Genetic Algorithm (GA) to maximize the function
$f(x) = x^2 - 1$
within the range [0, 31].
Each solution should be represented as a 5-bit binary string, since 25=322^5 = 3225=32, covering all possible values of x. The algorithm should start by generating a random population of binary-encoded values. Each individual in the population should be evaluated based on its fitness, which is determined by computing $f(x) = x^2 - 1$. The algorithm should then apply natural selection, retaining only the fittest individuals based on their fitness scores. Next, selected parents should undergo single-point crossover, where a random point in their binary representation is chosen, and their bits are swapped to create new offspring. To introduce genetic diversity, each offspring should undergo single-point mutation, where a randomly chosen bit is flipped ($0 \rightarrow 1$ or $1 \rightarrow 0$) with a small probability. The process should continue for a fixed number of generations or until the population converges to an optimal solution. The algorithm should finally return the best x value found along with its corresponding f(x) value, demonstrating how GA effectively optimizes the function over successive generations

National University of Computer and Emerging Sciences

Artificial Intelligence (AL-2002)

Lab Mid Exam

Total Time (hr): 1.5
Total Points: 50
Total Questions: 03

Paper B

Date: March 18th, 2025

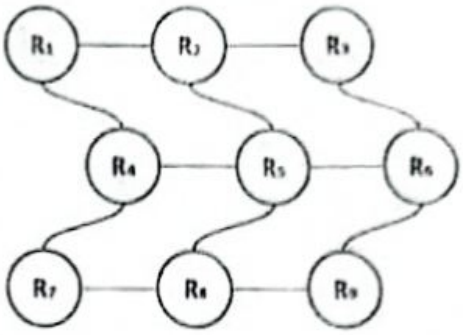
Course Instructor(s)

Mr. Talha Shahid. Miss. Alishba Subhani

Student Name _____ Roll No _____ Section _____ Student Signature _____

Question # 01 (LLO: 1) **[18 Points, 8 Weightage]**

In a **warehouse automation system**, robots are tasked with retrieving packages from storage racks. The storage racks are represented as an unweighted graph where each rack is a node, and accessible paths between racks are edges. A robot perceives its environment by detecting nearby racks and available paths. The robot's goal is to find the shortest path to a specified rack using **Iterative Deepening Depth-First Search (IDDFS).**



Robot's Starting Rack: **R1** | Robot's Goal: **R8**

Tasks:

1. Define the architecture of a goal-based robot agent and an environment class representing the warehouse layout. **[07]**
2. Implement the **Iterative Deepening Depth-First Search (IDDFS)** function within the agent class to find the shortest path to the goal rack. **[08]**
3. Create a `run_agent()` function to simulate the robot's behavior in the warehouse and display the shortest path found. **[03]**

Question # 02 (LLO: 3) **[18 Points, 9 Weightage]**

In a **treasure hunting game**, a player must navigate an island to find a hidden treasure. The island is represented as a grid where:

- . represents a traversable path.
- # represents an impassable obstacle.
- **S** represents the starting point (player's position).
- **T** represents the treasure's location.

The player can move up, down, left, or right. The player should use **Best-First Search (BFS)** with the **Euclidean distance** to the treasure as a heuristic to find the shortest path.

S	.	.	#	.	T
.	#	.	#	.	#
.	.	.	.	#	.
#	#	.	#	.	.
.	.	.	.	.	#

S = Starting Point
T = Treasure's Location
. = Traversable Path
# = Obstacle

Task:

1. Define the heuristic function which calculates the **Euclidean distance** between a point and the treasure. **[03]**
2. Define a **`generate_neighbours`** function to provide all possible neighbours of a given state based on the player's allowed movements. **[04]**
3. Implement the Best-First Search (BFS) function to find the shortest path to the treasure using the Euclidean distance heuristic. **[08]**
4. Create a **`main()`** function to execute the Best-First Search function, displaying the path taken and the total number of steps required. **[03]**

Question # 03 (LLO: 4) **[14 Points, 7 Weightage]**

Genetic Algorithm for the 0/1 Knapsack Problem | refer to the provided guide.

A logistics company is optimizing the space utilization of its delivery truck by selecting the best combination of packages to carry. Each package has a **weight** and **value**, and the truck has a **maximum weight limit**.

To solve this problem, you will implement a **Genetic Algorithm (GA)**, but with **modified selection, crossover, and mutation strategies**. Some parts of the algorithm are provided, and your task is to complete the missing functions.

You **must implement** the following functions to complete the GA:

1. **Generating Random Individuals** | `def create_random_individual(num_items):`
   - Create a random binary list where each gene represents whether an item is selected (1) or not (0).
2. **Initializing the Population** | `def initialize_population(pop_size, num_items):`
   - Generate an initial set of candidate solutions using the function from Step 1.
3. **Selecting Parents** | `def tournament_selection(population, fitness_scores):`
   - Instead of picking the top 50% directly, **implement tournament selection** where:
     - Randomly select **three individuals** from the population.
     - The individual with the **highest fitness** is chosen as a parent.
4. **Performing Two-Point Crossover** | `def crossover(parent1, parent2):`
   - Implement **two-point crossover**, where:
     - Select **two random crossover points**.
     - Swap the segment between them.

National University of Computer and Emerging Sciences

Artificial Intelligence Lab (AI1002)

Lab Mid Exam B

Total Time: 90 minutes
Total Points: 50
Total Questions: 03

Date: March 19th 2025

Course Instructor(s)
Ms Sarah , Mr Shafique, Mr. Muhammad Khalid,
Mr Abdullah Yaqoob

Semester: Spring 2025
Campus: Karachi
Department: CS, AI, SE, CY &DS

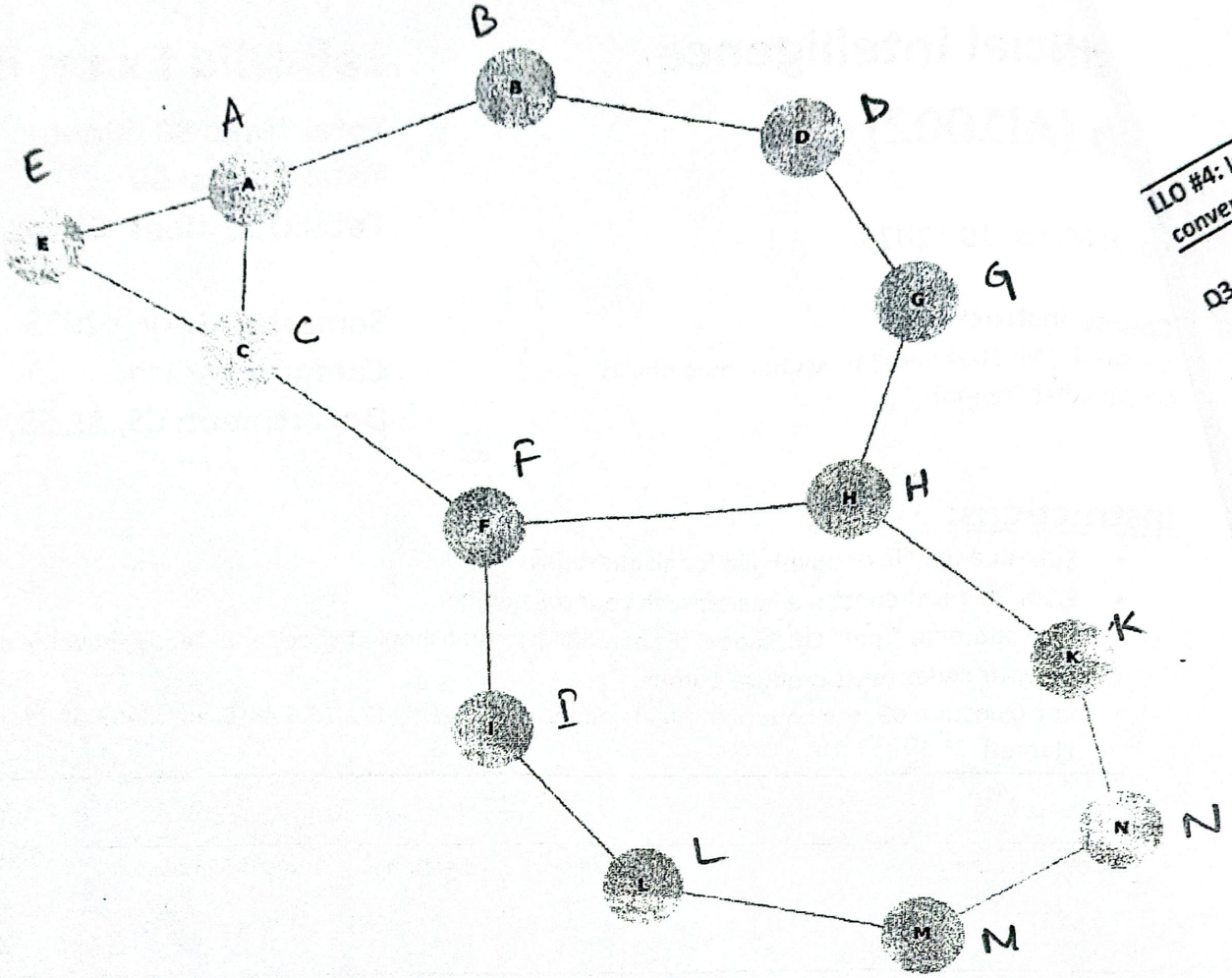
Instructions:

- Submit a .py file or .ipynb file for all questions.
- Each file must contain a header with your roll number.
- Also submit a ".pdf" OR ".docx" file containing screenshots of codes and clearly show the output.
- All your codes must produce output.
- For Question 03, the code is available on local storage (\\172.16.5.43\Exam Material\ (Your Teacher Name)).

Student Name	Roll No	Section	Student Signature

LLO # 1: Explain and implement BFS and DFS algorithms, comparing their efficiency and applications in uninformed search. ***[18 points]***

Q1 You are developing a **goal-based robot exploration agent** that needs to navigate through a graph. The graph is represented as an undirected graph, where each node represents a **room**, and the **edges** represent the doors connecting these rooms. The robot's goal is to navigate from a **starting room (E)** to a **target room (N)**. You need to use **Depth-First Search (DFS)** to solve this.



LLO #3: Apply and implement A* and GBFS algorithms, evaluating their heuristics, efficiency, and applications in informed search. ***[18 points]***

Q2. A drone is deployed in a 5×5 disaster relief zone to deliver emergency supplies to a survivor. The drone starts at (0,0) and must reach the survivor's location at (3,4) as quickly as possible using A search algorithm* with Manhattan distance as the heuristic function.

The disaster zone contains debris (represented as 1s in the grid), which the drone cannot fly through. The drone can move up, down, left, or right, but not diagonally.

Grid Representation:

```
grid = [
    [0, 0, 0, 0, 0],
    [0, 1, 1, 1, 0],
    [0, 0, 0, 1, 0],
    [0, 0, 0, 1, 0],
    [0, 1, 0, 0, 0],
]
```

LLO #4: Implement Genetic Algorithms to solve optimization problems, analyzing their operators, convergence, and performance. ***[14 points]***

Q3.

Instructions:

The code is available on local storage (\\172.16.5.43\Exam Material\ (Your Teacher Name)).

Problem Statement:

A delivery company has several delivery trucks and needs to find the optimal routes for each truck to deliver packages to multiple locations. Each truck has a limited capacity and each delivery point has a specific demand. The goal is to minimize the total travel distance while ensuring that each truck can carry the required amount of packages and that no truck exceeds its capacity.

Your task is to implement a Genetic Algorithm (GA) that will optimize the delivery routes for the trucks, considering the capacity constraints, delivery demands, and distance between delivery points.

Task:

1. By using Tournament Selection method (individual with the best fitness in this subset is chosen for reproduction), create a selection function. [4]
2. You are expected to write a crossover function using Two-Point crossover. [4]
3. You are expected to write a mutate function using Uniform Mutation (each gene in an individual has an equal probability of being randomly altered). [4]
4. Update such that the code runs completely and outputs best solution. [2]